

Advection Equation, 1D

$u(x, t)$

Models pure, rigid transport of physical quantity [e.g. concentration] that is being carried through space with zero diffusion, spreading or smearing involved, thus maintaining its initial shape as it travels.

OR in other words, any local change over time is purely caused by the surrounding flow pushing the spatial shape

Local Rate of change = $-\frac{\partial u}{\partial t}$ (Ambient Velocity) $\frac{\partial u}{\partial x}$ (Spatial Gradient)

OR $\frac{\partial u}{\partial t} + V \frac{\partial u}{\partial x} = 0$

Take note!

Spatial shape $\neq$ Spatial gradient

Why the negative?

Imagine a $+\frac{\partial u}{\partial x}$ [Peak conc. on Right]

Let $V > 0$ [Blob moving right]



From my perspective, the conc. will be decreasing since I'll see the blob from Right to Left [$\therefore \frac{\partial u}{\partial t} < 0$]

Characteristic Curve Derivation

Let $u(x, t) = c$ constant

$$\frac{d}{dt} u(x, t) = \frac{d}{dt} c$$

$$\frac{\partial u}{\partial x} \frac{dx}{dt} + \frac{\partial u}{\partial t} = 0$$

$$\frac{\partial u}{\partial x} \frac{dx}{dt} + [-V \frac{\partial u}{\partial x}] = 0 \quad [\text{using } \frac{\partial u}{\partial t} = -V \frac{\partial u}{\partial x}]$$

$$\frac{\partial u}{\partial x} \left[\frac{dx}{dt} - V \right] = 0$$

$$\frac{\partial u}{\partial x} = 0 \quad \text{or} \quad \boxed{\frac{dx}{dt} = V} \quad \text{--- } \textcircled{+}$$

[Reject $\therefore$ Trivial]

$\therefore$ This means the only way to guarantee that the conc. around you stays const. [$u(x, t) = c$] is if your personal speed matches the fluid's speed.

This makes sense cuz if $\frac{dx}{dt} = V$, the relative motion bwn me & the blob is zero. And since the blob spatial shape is constant, the amount of stuff around me will not change

Integrating Eqn $\textcircled{+} \implies x(t) = \underbrace{Vt}_{\text{Distance travelled}} + \underbrace{c}_{\text{Starting position}}$

Since $u(x, t) = \text{constant}$, in other words $u(x, t)$ is only dependent on that constant

$$u(x, t) = f(c)$$

$$u(x, t) = f(x - Vt)$$

This makes sense cuz if my personal speed ($\frac{dx}{dt}$) matches the Blob's speed (V), our relative motion = 0, thus the conc. ($u(x, t)$) will be constant. The only way to get a diff conc. is by starting at a diff point. And to calc. that starting point, c we just take our [current position, x - Distance travelled, Vt]

Further Explanations

Characteristic Curve $\implies$ General Path of constant concentration

$u(x, t) \implies$ concentration measured at one specific location (x) & time (t)

$f(x)$ or $g(x) \implies$ Reference shape [snapshot]

Connection btwn $u(x, t)$ and $f(x)$

$f(x)$ contains the complete set of every single concentration

reading across all space at one specific slice of time.

$u(x, t)$ simply calculates which specific reading from that

frozen slice ($g(x)$) is currently passing by a location, x at time, t

$$x(t) = c + Vt$$

↓ General version

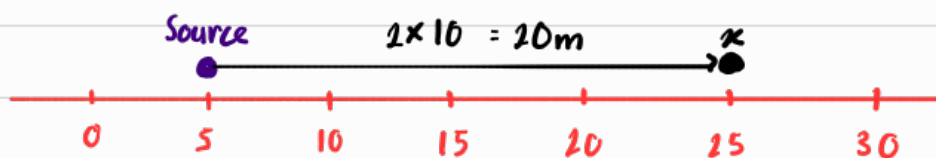
Current position = ^{Snapshot origin / Ref. position} Source Coordinate + Dist. travelled since $f(x)$ snapshot

The corresponding coordinate on reference shape

^{Snapshot origin}
Source Coordinate = Current position - Dist. travelled since $f(x)$ snapshot

Example

Let's say $V = 2$, time passed (t) = 10, current position (x) = 25



^{snapshot origin}
 $\therefore$ source coordinate = $25 - [2 \times 10]$
 $= 5$

Example 1 [Recommended Method]

Find the solution of the advection equation

$$\frac{\partial u}{\partial t} - 2 \frac{\partial u}{\partial x} = 0$$

with initial conditions $u(x, 0) = e^{-x^2}$ for $-\infty < x < \infty$

To find characteristic curve, we have to find

which paths is $u(x, t)$ a constant?

Let $u(x, t) = A$ ^{constant}

$$\Rightarrow \frac{d}{dt} u(x, t) = \frac{d}{dt} (A)$$

$$\frac{\partial u}{\partial x} \frac{\partial x}{\partial t} + \frac{\partial u}{\partial t} = 0$$

$$\frac{\partial u}{\partial x} \frac{\partial x}{\partial t} + \left[2 \frac{\partial u}{\partial x} \right] = 0$$

$$\frac{\partial u}{\partial x} \left[\frac{\partial x}{\partial t} + 2 \right] = 0$$

$$\frac{\partial u}{\partial x} = 0 \quad \text{or} \quad \frac{\partial x}{\partial t} = -2$$

[Reject
 $\therefore$ Trivial]

$$x(t) = -2t + C, \quad C \in \mathbb{R}$$

general path where $u(x, t)$ is constant

Since...

1) every general path is identified by its starting coordinate, C

2) $u(x, t)$ depends on the specific path it takes

$$\Rightarrow u(x, t) = f(C)$$

$$\text{Using eqn } \textcircled{1} \Rightarrow C = x + 2t$$

$$\therefore u(x, t) = f(x + 2t) \quad [\text{General Solution}]$$

By using initial condition, $u(x, 0) = e^{-x^2}$

$$f(x + 2(0)) = e^{-x^2}$$

$$f(x) = e^{-x^2}$$

$$\therefore u(x, t) = f(x + 2t)$$

$$u(x, t) = e^{-(x+2t)^2} \quad \text{ff}$$

Alternative method

Since we know for $u(x, t)$ to be const.

our tracking speed, $\frac{\partial x}{\partial t}$ must equals the fluid's velocity, V .

$$\text{By comparing to } \frac{\partial u}{\partial t} + V \frac{\partial u}{\partial x} = 0,$$

$$\Rightarrow V = -2$$

$$\therefore \frac{\partial x}{\partial t} = -2 \quad \text{ff}$$

Alternate Method

$$\text{Using } u(x, t) = f(x - Vt)$$

$$= f(x - (-2)t)$$

$$u(x, t) = f(x + 2t)$$

Example 1 [Understanding Method]

Find the **solution** of the advection equation

$$\frac{\partial u}{\partial t} - 2 \frac{\partial u}{\partial x} = 0$$

with initial conditions $u(x, 0) = e^{-x^2}$ for $-\infty < x < \infty$

Let $f(x)$ be the reference shape function at $t = 0$

$$f(x) = u(x, 0)$$

$$f(x) = e^{-x^2}$$

By comparison $\bar{u} \quad \frac{\partial u}{\partial t} + V \frac{\partial u}{\partial x} = 0 \Rightarrow V = -2$

Since the spatial shape is constant, the conc. of any point at any time can be found by "rewinding" & finding the conc. of that point in the known **Reference Shape function** spatial shape at that given time

$$\begin{aligned} \therefore \text{Reference position} &= \text{Current position} - \text{Distance travelled since Reference shape} \\ &= x - V(t - 0) \\ &= x - Vt \\ &= x - (-2)t \\ &= x + 2t \end{aligned}$$

$$\therefore u(x, t) = f(x + 2t)$$

$$u(x, t) = e^{-(x+2t)^2} \quad \#$$

Example 1a [Just for understanding sake]

Find the solution of the advection equation

$$\frac{\partial u}{\partial t} - 2 \frac{\partial u}{\partial x} = 0$$

with initial conditions $u(x, 2) = e^{-x^2}$ for $-\infty < x < \infty$

Let $f(x)$ be the reference shape function at $t=2$

$$f(x) = u(x, 2)$$

$$f(x) = e^{-x^2}$$

By comparison with $\frac{\partial u}{\partial t} + V \frac{\partial u}{\partial x} = 0 \Rightarrow V = -2$

Since the spatial shape is constant, the conc. of any point at any time can be found by "rewinding" & finding the conc. of that point in the known
 Reference Shape function
 spatial shape at that given time

$$\begin{aligned} \therefore \text{Reference position} &= \text{Current position} - \text{Distance travelled since Reference shape} \\ &= x - V(t - 2) \\ &= x - (-2)(t - 2) \\ &= x + 2t - 4 \end{aligned}$$

$$\therefore u(x, t) = f(x + 2t - 4)$$

$$u(x, t) = e^{-(x + 2t - 4)^2} \quad \checkmark$$

Wave Equation, 1D

For **advection**, we were interested in **things that only move due to its surroundings and not itself**. [e.g. dye]

Thus, since the thing can't move itself, the **local change over time** is purely caused by the **surrounding flow** pushing the **spatial gradient** past your position.

For **wave equation**, we are interested in **things that only move themselves without the help of its surroundings** [e.g. string releasing tension]

Thus, in this situation the **local acceleration over time** is purely driven by how sharply the physical shape is bent, scaled by the string's internal "snap-back multiplier" c^2 .

Decides how much acceleration u get for a given ammount of bending [Tightness of string]

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

Local acceleration = wave speed squared = Spatial Curvature

$$\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0$$

D'Alembert's Solution [No Boundaries]

From $\frac{\partial^2 u}{\partial t^2} - c^2 \frac{\partial^2 u}{\partial x^2} = 0$,

Using $a^2 - b^2 = (a+b)(a-b)$
 $\Rightarrow \left[\frac{\partial}{\partial t} + c \frac{\partial}{\partial x} \right] \left[\frac{\partial}{\partial t} - c \frac{\partial}{\partial x} \right] u = 0$

or treating derivatives as instructions not values

$$\Rightarrow \left[\frac{\partial u}{\partial t} + c \frac{\partial u}{\partial x} \right] \left[\frac{\partial u}{\partial t} - c \frac{\partial u}{\partial x} \right] = 0$$

Advection Egn
flowing right

Advection Egn
flowing left

Because of linear superposition,
this means the Wave Egn is just
1 Advection overlapping.

Reference Shape 1 Reference shape 2

$$\therefore u(x, t) = F(x+ct) + G(x-ct)$$

Example 1 $\frac{\partial u}{\partial t} \Big|_{t=0} = 0$

$$u(x, 0) = \sin(x)$$

A string is initially held at rest with a shape of the sine curve $\sin(x)$ before being released for $t > 0$. Use D'Alembert's solution to find the solution to the wave equation given these initial conditions.

General solution : $u(x, t) = F(x+ct) + G(x-ct)$

$$\frac{\partial u}{\partial t} = c F'(x+ct) - c G'(x-ct)$$

Using condition $u(x, 0) = \sin(x)$

$$\Rightarrow F(x+c(0)) + G(x-c(0)) = \sin(x)$$

$$F(x) + G(x) = \sin(x) \quad \text{--- (1)}$$

Using condition $\frac{\partial u}{\partial t} \Big|_{t=0} = 0$

$$\Rightarrow c F'(x+c(0)) - c G'(x-c(0)) = 0$$

$$F'(x) - G'(x) = 0$$

$$F'(x) = G'(x)$$

Integrating both sides,

$$F(x) = G(x) + k \quad \text{--- (2)}$$

(2) $\rightarrow$ (1)

$$[G(x) + k] + G(x) = \sin(x)$$

$$2G(x) + k = \sin(x)$$

$$G(x) = \frac{1}{2} \sin(x) - \frac{k}{2}$$

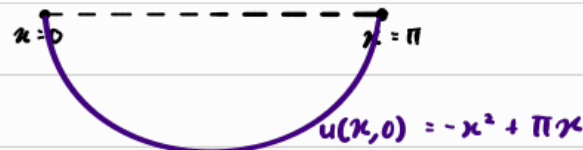
$$\therefore F(x) = \left[\frac{1}{2} \sin(x) - \frac{k}{2} \right] + k$$

$$F(x) = \frac{1}{2} \sin(x) + \frac{k}{2}$$

$$\therefore u(x, t) = \left[\frac{1}{2} \sin(x+ct) + \frac{k}{2} \right] + \left[\frac{1}{2} \sin(x-ct) - \frac{k}{2} \right]$$
$$= \frac{1}{2} \sin(x+ct) + \frac{1}{2} \sin(x-ct)$$

Separation of Variables

[Given Boundaries]



Example 2

Find the solution of the wave equation on a string with a length of π units

$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

subject to the boundary conditions

$$u(0, t) = 0 \text{ and } u(\pi, t) = 0 \text{ for } t > 0$$

$$X(0) = 0 \quad X(\pi) = 0$$

and the initial conditions

$$u(x, 0) = x(\pi - x) \text{ and } \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0 \text{ for } 0 < x < \pi$$

Conditions given

$$u(0, t) = 0 \Rightarrow X(0) = 0$$

$$u(\pi, t) = 0 \Rightarrow X(\pi) = 0$$

$$u(x, 0) = x(\pi - x)$$

$$\left. \frac{\partial u}{\partial t} \right|_{t=0} = 0 \Rightarrow X(x) T'(0) = 0$$

$$T'(0) = 0$$

Assume $u(x, t) = X(x) \cdot T(t)$

Using original PDE, $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$

$$\Rightarrow X T'' = c^2 X'' T$$
$$\frac{X''}{X} = \frac{T''}{c^2 T} = \lambda$$

$$\therefore X'' - \lambda X = 0 \quad \text{--- ①}$$

$$T'' - c^2 \lambda T = 0 \quad \text{--- ②}$$

Solving for ①

Characteristic Eqn: $r^2 - \lambda = 0$

$$r = \pm \sqrt{\lambda}$$

Case 1, $\lambda > 0$

Let $\lambda = \omega^2$

$$\therefore \text{GS: } X(x) = C_1 e^{\omega x} + C_2 e^{-\omega x}$$

Applying $X(0) = 0 \Rightarrow C_2 = -C_1$

$$\therefore X(x) = C_1 [e^{\omega x} - e^{-\omega x}]$$

Applying $X(\pi) = 0 \Rightarrow C_1 [e^{\omega \pi} - e^{-\omega \pi}] = 0$

$$\therefore C_1 = 0 \Rightarrow C_2 = 0$$

[Trivial]

Case 2, $\lambda = 0$

$$X'' = 0$$

$$X(x) = Ax + B$$

Applying $X(0) = 0, X(\pi) = 0$

$$\Rightarrow A = 0, B = 0$$

[Trivial]

Case 3, $\lambda < 0$

Let $\lambda = -(\omega^2)$

$$\therefore \text{GS: } X(x) = A \cos(\omega x) + B \sin(\omega x)$$

Applying $X(0) = 0 \Rightarrow A = 0$

$$\therefore X(x) = B \sin(\omega x)$$

Applying $X(\pi) = 0 \Rightarrow B \sin(\omega \pi) = 0$

$$B = 0 \text{ or } \sin(\omega \pi) = 0$$

[Reject] [Trivial]

$$\omega = n, \text{ where } n = 1, 2, 3, \dots$$

$$\therefore X_n(x) = B \sin(n x)$$

As expected since we have 2 roots, ^{$x=0, x=\pi$} only the λ that give trigo is valid

Solving for ②

$$T'' - c^2 \lambda T = 0$$

$$\text{Since } \lambda = -(\omega^2) = -n^2$$

$$\therefore T'' + (cn)^2 T = 0$$

$$\text{Characteristic Eqn: } r^2 + (cn)^2 = 0$$

$$r = \pm \sqrt{-(cn)^2}$$

$$= \pm i(cn)$$

$$\therefore \text{GS: } T_n(t) = C \cos(cnt) + D \sin(cnt)$$

$$\text{Using condition } \left. \frac{\partial u}{\partial t} \right|_{t=0} = 0 \Rightarrow T'(0) = 0$$

$$\Rightarrow T'_n(t) = -C(cn) \sin(cnt) + D(cn) \cos(cnt)$$

$$T'_n(0) = 0 + D(cn)(1)$$

$$0 = D(cn)$$

$$D = 0$$

$$\therefore T_n(t) = C \cos(cnt)$$

$$\therefore u_n(x, t) = X_n(x) \cdot T_n(t)$$

$$= b_n \sin(nx) \cos(cnt)$$

$$u(x, t) = \sum_{n=1}^{\infty} b_n \sin(nx) \cos(cnt) \quad [\text{General solution}]$$

$$\text{Using condition } u(x, 0) = x(\pi - x)$$

$$\Rightarrow u(x, 0) = \sum_{n=1}^{\infty} b_n \sin(nx)$$

$$\sum_{n=1}^{\infty} b_n \sin(nx) = x(\pi - x)$$

Using Fourier formula,

$$b_n = \frac{2}{\pi} \int_0^{\pi} x(\pi - x) \sin(nx) dx$$

:

$$b_n = \frac{4}{\pi n^3} [1 - (-1)^n], \text{ for } n = 1, 2, 3, \dots$$

$$b_n = \frac{8}{\pi n^3}, \text{ for } n = 1, 3, 5, \dots$$

$$u(x, t) = \sum_{n=1,3,5}^{\infty} \frac{8}{\pi n^3} \sin(nx) \cos(cnt) \quad \neq$$

